Course Information

Course Number:	221
Course Title:	Data Structures and Algorithms: CSCE 221
Section:	551
Time:	Lectures: MWF 9:10AM-10:00AM Labs: MW 12:40PM-1:30PM
Location:	Lectures: CLB 114 Labs: MAIN 303
Credit Hours:	4

Instructor Details

Instructor:	Ünal Göktaş
Office:	SeaAggie Center 706
Phone:	409-740-5515
E-Mail:	ugoktas@tamu.edu
Office Hours:	MW 10:30AM-11:30AM or by appointment

TA:	TBA on Canvas Course Pages
E-Mail:	TBA on Canvas Course Pages
Office Hours:	TBA on Canvas Course Pages

Course Description and Prerequisites

Specification, analysis, and implementation of abstract data types for lists, stacks, queues, trees, hash tables, graphs, and their associated algorithms. Performance trade-offs of different implementations; asymptotic analysis of running time and memory usage. Includes the execution of students' programs written in C++; emphasis on adherence to good software engineering principles.
Prerequisites: CSCE 120 or CSCE 121
Corequisites: CSCE 222 or ECEN 222

Course Learning Outcomes

At the end of this course, students should be able to:

- Design and implement diverse data structures that allow easy access and manipulation of data using C++ programming language.
- Apply the Big-O asymptotic notation to analyze and select an efficient algorithm for solving a given problem with respect to runtime and memory usage.
- Identify the latest developments in the Computer Science area or be familiar with Turing award winners.

Textbook and/or Resource Materials

- **Required Textbook:** "*Data Structures and Algorithm Analysis in C++*", 4th Edition, Mark A. Weiss, 2014, Pearson, ISBN-13: 978-0132847377 or ISBN-10: 013284737X. Also, you may need a textbook on C++ language from the previous semester, or any good C++ reference book(s).

- **Required: Web, Zoom, Canvas, and Gradescope access.** The course material, assignments turn-in, and posting grades are handled through Canvas and/or Gradescope. The recommended browsers to access Canvas are Mozilla Firefox, Google Chrome, and Microsoft Edge.

Grading Policy

Grading Criteria	
Homework Assignments	9%
Programming Assignments	27%
Culture Assignment	4%
Quizzes	15%
Exam 1	15%
Exam 2	15%
Exam 3	15%

Grading Scale	
90–100	A
80–89.9	B
70–79.9	C
60–69.9	D
0–59.9	F

- Your final grade will be determined based on written homework, programming assignments, cultural assignment, quizzes, and exams.
- The assignment handout will provide point distribution. Each programming assignment will be graded focused on: algorithm design, usage of data structures and/or new user-defined types and their implementation, its correctness, tests, a report describing implemented algorithms and data structures, and results of computational experiments.
- **All grade appeals must be made no later than one week after the grade is posted.**

Graded Attendance

- The students' lab attendance is required and will be taken on a regular basis.
- The lecture attendance will be taken by pop quizzes. The students are responsible to learn all material covered in class, reading the assigned text from the textbook, and do homework assignments.
- 2 bonus points will be added to your final score if you have perfect lab attendance, or 1 point if you have only one lab absence.
- Make-up exams and quizzes will only be given with documented University-approved excuses, see University Regulations. Student Rule 7 explains policies and excused absences. Please discuss unusual circumstances in advance with the instructor.

Notes about graded materials

- **All homework assignments will be announced in class and posted on Canvas.** See the link to the course calendar on Canvas for deadlines.
- The written part of homework assignments or programming reports, and the Cover Page should be typed **preferably using LaTeX or LaTeX-based apps**: overleaf (an online LaTeX text editor) or LyX (document processor based on LaTeX), see the class webpage for a LyX tutorial. The homework **in PDF format** should be submitted to Canvas.

- All programming assignments **should be implemented in C++**, compiled and run on a computer (preferably using Linux), and transferred to Gradescope for grading.
- The Culture Assignment allows you to explore the latest developments in Computer Science and Engineering or learn about famous computer scientists like Turing Award winners. The assignments should be submitted to Canvas.
- Quizzes are over material covered during the lectures and assigned reading from the textbook.
- The Instructor and TAs can request oral exams to check students' knowledge of the material covered by tests or assignments.

Late Work Policy

A homework assignment will be accepted up to 1 day late with a 15% penalty. Once solutions have been discussed or handed out the assignments will not be accepted. Please discuss unusual circumstances in advance with the instructor. Work submitted by a student as makeup work for an excused absence is not considered late work and is exempted from the late work policy (see Rule 7).

Course Schedule

Module	Week	Day	Topic
Module 1	Jan. 16	T	Introduction to Data Structures Pointers, Arrays, Vectors Introduction to Analysis of Algorithms
Module 2	Jan. 22	M	Complexity of Search and Sort Algorithms Algorithms – Best, Worst, Average Cases The Lower Bound Theorem
Module 3	Jan. 29	M	Linear Sort Algorithms Stack ADT – Array Based Implementation Amortized Analysis
Module 4	Feb. 5	M	Queue ADT – Array-based Implementation Stack and Queue Applications. Parser
Module 5	Feb. 12	M	Linked Lists in C++ Linked List-based Stack and Queue **Exam 1**
Module 5	Feb. 19	M	Linked List-based Stack and Queue Dequeues
Module 6	Feb. 26	M	Recursive Algorithms Analysis of Divide-and-Conquer Algorithms
Module 7	Mar. 4	W	Iterating and Master Methods Trees | Binary Search Trees
Module 8	Mar. 18	M	AVL Trees Red-Black Trees 2–4 Trees **Exam 2**

Module 9	Mar. 25	M	Skip Lists | Hashing	
Module 10	Ap1. 1	M	Heap | Huffman Trees Priority Queues	
Module 11	Apr. 8	M	Graphs | BFS | DFS Digraph Algorithms	
Module 12	Apr. 15	M	Shortest Path Algorithms	The topics and related chapters
Module 12	Apr. 22	M	Dijkstra's Algorithm Minimum Spanning Trees Prim's Algorithm Disjoint sets ADT	
Module 13	Apr. 29	M	Kruskal's Algorithm	
	May. 3		**Final Exam:** **Section 551, 5/3/2024, 8:00AM-10:00AM**	

from the textbook:

Introduction	
C++ Overview: Arrays, Vectors, Strings, Matrices	Chap. 1
Introduction to Analysis of Algorithms	Chap. 2
Lists, Stacks, and Queues	Chap. 3
Trees and Search Trees	Chap. 4
Balanced Search Trees	Chap. 12.2
Hashing	Chap. 5
Priority Queues. Heaps.	Chap. 6
Sorting	Chap. 7
Disjoint Sets	Chap. 8
Graphs	Chap. 9
Algorithms Design	Chap. 10

All changes in the schedule will be announced in class and on the class calendar. Please see also University Academic Calendar.
Use Canvas to get access to the lecture notes, quizzes and tests, and assignments, and check your grades on a regular basis during the semester.

Technology Support - Texas A&M University at Galveston students can direct their technical questions to helpdesk@tamug.edu

Learning Resources – On-campus learning resources to support students in achieving excellence are available through the Center for Academic Learning Support (tamug.edu/cals).

University Policies

This section outlines the university level policies that must be included in each course syllabus. The TAMU Faculty Senate established the wording of these policies.

Attendance and Makeup Work Policies

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to Student Rule 7 in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" (Student Rule 7, Section 7.4.1).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" (Student Rule 7, Section 7.4.2).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See Student Rule 24.)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" (Section 20.1.2.3, Student Rule 20).

You can learn more about the Honor Council Rules and Procedures as well as your rights and responsibilities at tamug.edu/HonorSystem.

No Use of Generative AI Tools Permitted

All assignments should be fully prepared by the student. Developing strong competencies in the skills associated with this course, from student-based brainstorming to project development, will prepare you for success in your degree pathway and, ultimately, a competitive career. Therefore, the use of generative AI tools to complete any aspect of assignments for this course are not permitted and will be treated as plagiarism. If you have questions about what constitutes a violation of this statement, please contact me.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office located in the Student Services Building or at (409) 740-4587 or visit tamug.edu/counsel/Disabilities. Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see University Rule 08.01.01.M1):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with the Counseling Office in the Seibel Student Center, or call (409)740-4587. For additional information, visit tamug.edu/counsel.

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the Galveston Campus' Title IX webpage.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care by utilizing available resources and services on your campus.

Students who need someone to talk to can call (409) 740-4736 from 8:00 a.m. to 5:00 p.m. weekdays or visit tamug.edu/counsel for more information. For 24-hour emergency assistance during nights and weekends, contact the TAMUG Police Dept at (409) 740-4545. 24-hour emergency help is also available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.

COVID Statement

To help protect Aggieland by the Sea and stop the spread of COVID-19, Texas A&M University at Galveston urges students to be vaccinated and to wear masks in classrooms and all other academic facilities on campus, including labs. Doing so exemplifies the Aggie Core Values of respect, leadership,

integrity, and selfless service by putting community concerns above individual preferences. COVID-19 vaccines and masking — regardless of vaccination status — have been shown to be safe and effective at reducing spread to others, infection, hospitalization, and death.

Texas A&M at Galveston - Campus-Specific Policies

Classroom Access and Inclusion Statement

Texas A&M University is committed to engaged student participation in all of its programs and courses and provides an accessible academic environment for all students. This means that our classrooms, our virtual spaces, our practices and our interactions are as inclusive as possible and we work to provide a welcoming instructional climate and equal learning opportunities for everyone. If you have an instructional need, please notify me as soon as possible.

The Aggie Core values of respect, excellence, leadership, loyalty, integrity and selfless service in addition to civility, and the ability to listen and to observe others are the foundation of a welcoming instructional climate. Active, thoughtful and respectful participation in all aspects of the course supports a more inclusive classroom environment as well as our mutual responsibilities to the campus community.

Statement on the Family Educational Rights and Privacy Act (FERPA)

FERPA is a federal law designed to protect the privacy of educational records by limiting access to these records, to establish the right of students to inspect and review their educational records and to provide guidelines for the correction of inaccurate and misleading data through informal and formal hearings. Currently enrolled students wishing to withhold any or all directory information items may do so by going to howdy.tamu.edu and clicking on the "Directory Hold Information" link in the Student Records channel on the MyRecord tab. The complete FERPA Notice to Students and the student records policy is available on the Office of the Registrar webpage.

Items that can never be identified as public information are a student's social security number, citizenship, gender, grades, GPR or class schedule. All efforts will be made in this class to protect your privacy and to ensure confidential treatment of information associated with or generated by your participation in the class.

Directory items include name, UIN, local address, permanent address, email address, local telephone number, permanent telephone number, dates of attendance, program of study (college, major, campus), classification, previous institutions attended, degrees honors and awards received, participation in officially recognized activities and sports, medical residence location and medical residence specialization.

Statement on Course Evaluations

Texas A&M will use an online course evaluation tool through AEFIS (Assessment, Evaluation, Feedback & Intervention System). We highly encourage you to complete an evaluation for each course on your schedule. Student input is a critical component used to improve curriculum and teaching. Each faculty member values your input to improve his/her methodology. Your comments can also significantly impact the mix and membership of faculty. The course evaluations will be available through

TAMU.AEFIS.NET and you will receive an email notification with a direct link to the system when the course evaluation becomes available.

Department Policies

Departmental units may establish their own policies and minimum syllabus requirements. As long as these policies and requirements do not contradict the university level and college level requirements, departments can add them in this section. Please remove this section if not needed.